

CSMVessel0000000005 Cruise Ship Resilience_ Entertainment Systems

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Maritime / Entertainment

Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
GaN audio amplifiers	Not specified	GaN Class-D: 97% efficiency (same as CSMFAB005 technology)	V1.0 had no specific amplifier specification for retrofit
LRAM theater acoustics	Not in V1.0	MIT 2025 LRAM at theater scale: -65 dB at 250 Hz in 15mm panel	Theater acoustic isolation from ship vibration unaddressed in V1.0
Casino slot machines	General EMP vulnerability	V2.0: BFRP case + MXene shield: \$445/machine; 500 machines = \$222,500/ship	V1.0 had no per-machine protection cost estimate
EMI enclo SE	V1.0 copper cage SE=45 dB	MXene SE=92 dB in BFRP enclosures	47 dB shielding deficit in all AV server rooms
Signal wiring	Copper AV cables assumed	PMMA POF optical fiber signal bus (CSMFAB013 marine adaptation)	Metallic AV signal cables form GIC antennas in ship — structural failure of entertainment

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.